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Part 1



Introduction

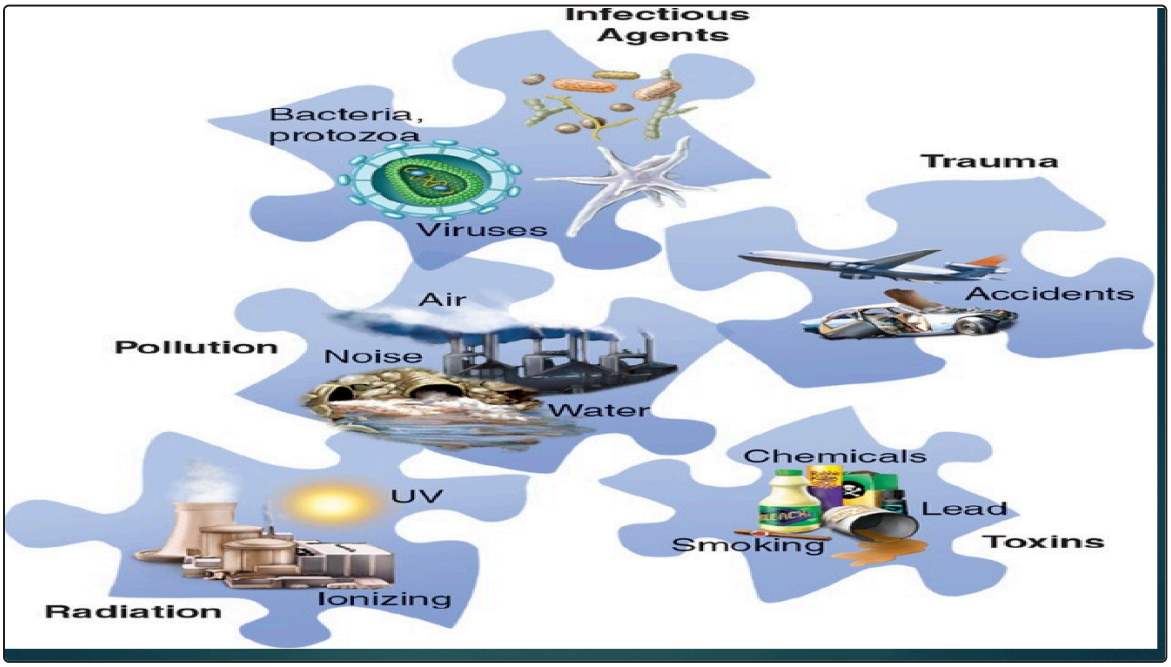
- Environmental Health and Toxicology is responsible for two distinct areas of **public health concern** as follows:-
- **Environmental health:-** Relationship between people and their environment Promotes Human health and wellbeing.
- Fosters healthy and safe communities
- **Toxicology:-** study the adverse effects of chemical, physical, and biological agents.
- field of science that helps us understand the harmful effects that chemicals, substances, or situations, can have on people, animals, and the environment.
- **Major environmental health concern:-**
- Health Impacts
- Air Quality
- Water and Sanitation



- **Chemical:** synthetic chemicals such as pesticides, disinfectants, and pharmaceuticals. Harmful natural chemicals also exist, including Urushiol(an oil in poison ivy, etc.)
- **Biological:** biological substances from ecological interactions. Viruses, bacteria, and other pathogens. Infectious (communicable and transmissible) diseases, parasites, we can reduce the likelihood of infection through vaccination and immunizations, keeping a distance from sick/infected people, etc.
- **Cultural**: results from the place we live, our socio-economic status, our occupation, and our behavioral choices.
- **Psychosocial:** due to stress, hostility, depression, and hopelessness. This affects the mental well-being of a person.

Infectious organisms

- otherwise **called pathogens or disease-causing organisms**, they are classified into five. Bacteria, viruses, Fungi, Protozoa, helminthes, and worms
- **Morbidity and quality of life:** any physical or psychological state considered to be outside the realm of normal well-being. The term **morbidity** is often used to **describe illness, impairment, or degradation of health, especially when discussing chronic and age-related diseases**.
- Morbidity doesn't necessarily mean that your illness is immediately life-threatening. Over time, however, if an illness continues, it may increase your **risk of mortality (death)**. The most common are **Heart disease, cancer, chronic lower respiratory diseases, stroke, Alzheimer's disease, diabetes mellitus, pneumonia and influenza, kidney disease, and suicide**, accounting for almost **75 percent of deaths in the U.S in 2013**. In addition to infectious diseases, **foodborne illness, associated infections, and sexually transmitted diseases also contribute to higher morbidity among Americans.**



RE-EMERGING INFECTIOUS DISEASES

Disease	Infectious agent	Contributing factors
Cryptosporidiosis	*Cryptosporidium parvum*	International travel; contaminated water supplies
Diphtheria	*Corynebacterium diphtheriae*	Interruption of immunization program due to political changes
Influenza	Influenza virus	Genetic re-assortment
Malaria	*Plasmodium* species	Drug resistance; inadequate mosquito control
Pertussis	*Bordetella pertussis*	Refusal to vaccinate; decreased vaccine efficiency; waning immunity in adults
Rabies	Rhabdovirus	Breakdown in public health measures; travel; changes in land use
Tuberculosis	*Mycobacterium tuberculosis*	Antibiotic resistance; increased immunocompromised populations
Yellow fever	Flavivirus	Urbanization; insecticide resistance

Table 8.2 Microbiology: A Clinical Approach (© Garland Science)

Ecological diseases

- Emerging ecological diseases present a major threat to world health. On average, two new species of the human virus are reported each year, most having an animal origin. Recent examples are **SARS**, **swine flu**, and **avian influenza**. In 2014, the **Ebola virus** re-emerged from a bat reservoir, causing a major epidemic.
- Climate change could lead to ***Aedes* mosquitoes** establishing in New Zealand, and with them **dengue** and **Zika viruses**.
- The problem of disease emergence also extends to animal and plant populations. In most current epidemics, ecological factors (e.g., **migration, climate, and agricultural practices**) play a more significant role in disease emergence than evolutionary changes in pathogens or hosts. Specifically, factors affecting the environment include the depletion of forests, the expansion and modernization of agricultural practices, and natural disasters such as floods.





Natural Toxic Chemicals

Many toxic chemicals occur in nature. For example, plants produce toxic chemicals to protect themselves from pests. Animals produce toxins for protection and to capture prey. In other cases, toxic chemicals are simply a by-product of metabolism.
Some **natural elements** and **minerals are poisonous.** Here are some examples of **natural toxic chemicals**

- **Mercury**
- **Snake venom**
- **Caffeine in coffee, tea, kola and cocoa**
- **Arsenic**
- **Ricin from castor beans**
- **Petroleum**
- **Hydrogen sulfide**
- **Chlorine gas**
- **Smoke**

Industrial and Occupational Toxic Chemicals

The **US Occupational Safety and Health Administration** (OSHA) have identified several chemicals it considers highly hazardous and toxic. Some of these are laboratory reagents, while others are used commonly in certain industries and trades. Certain pure **elements** are included. Here are a few substances on the list (which is extremely long):

- Acetaldehyde
- Acetone
- Acrolein
- **Bromine**
- **Chlorine**
- Cyanogen
- Isopropyl alcohol
- l-limonene
- Hydrogen peroxide >35%



A healthy diet is essential for good health

- It protects you against many chronic non-communicable diseases, such as heart disease, diabetes, and cancer. Eating a variety of foods and consuming less salt, sugars, saturated and industrially-produced trans-fats, are essential for a healthy diet. A healthy diet comprises a combination of different foods. These include:
- Staples like cereals (wheat, barley, rye, maize, or rice) or starchy tubers and roots (potato, yam, taro, or cassava).
- Legumes (lentils and beans).
- Fruit and vegetables.
- Foods from animal sources (meat, fish, eggs, and milk).

References

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